

# Network Science

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## Lecture 08: Network Dynamics

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# From Structure to Dynamics

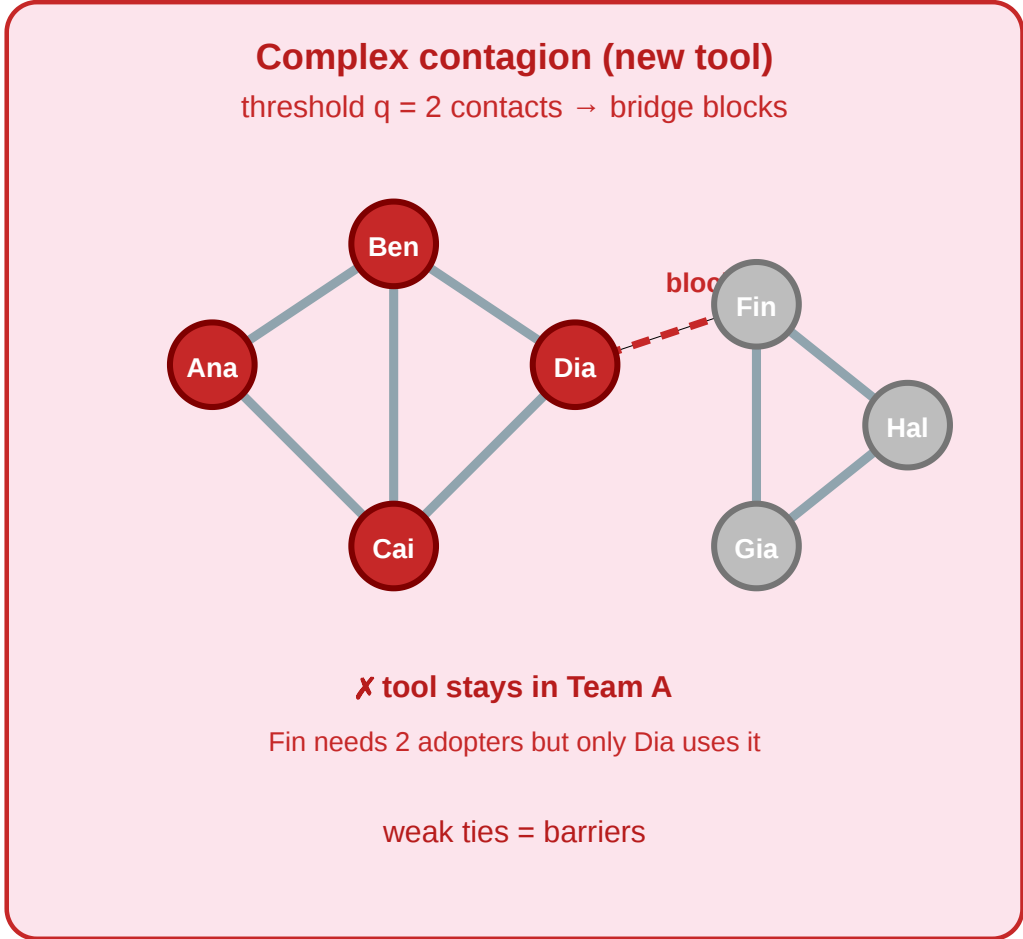
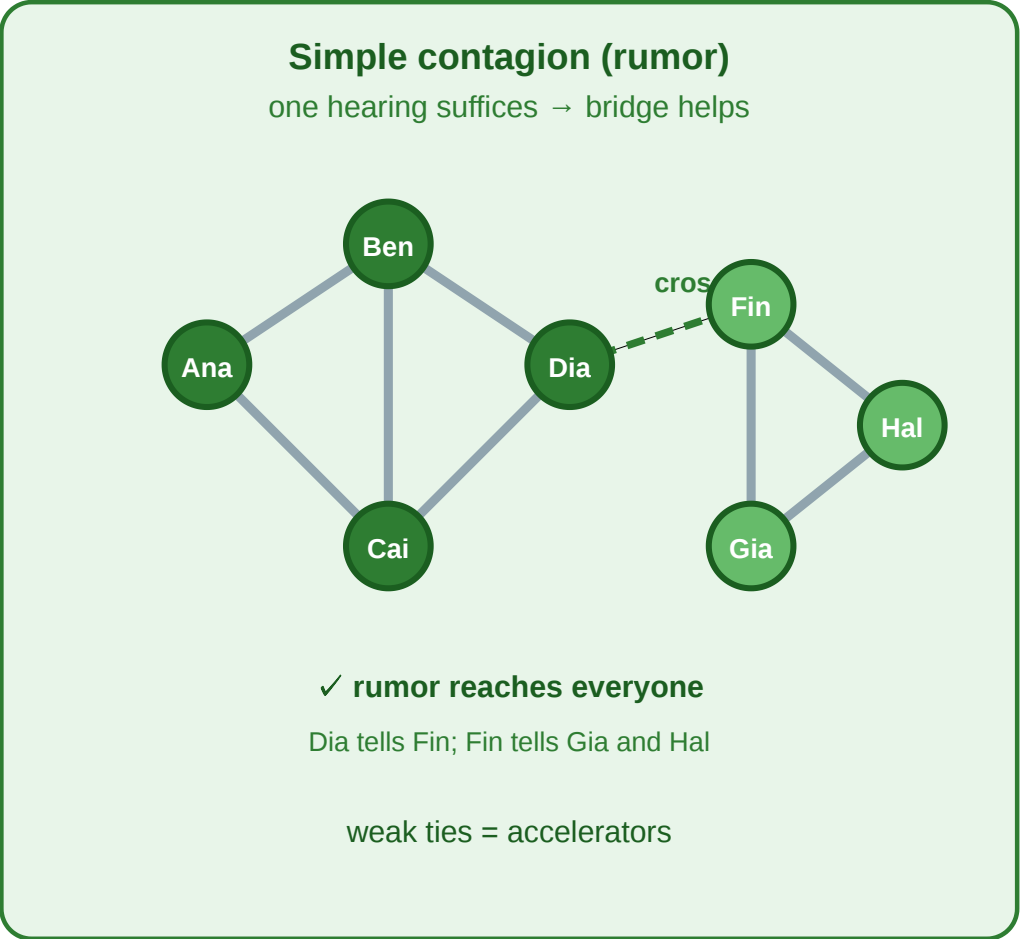
Lectures 03–07 analyzed networks as static objects: edges, roles, communities, balance, short paths. Today we ask: **what happens *on* the network?**

The same structural features we studied — weak ties, bridges, communities, hubs — play different roles depending on *what is spreading*. Let us return to the workplace one last time.

# A Final Visit to the Workplace

## Same network, two processes — opposite outcomes

A rumor (simple contagion) crosses the bridge; a tool adoption (complex contagion) does not



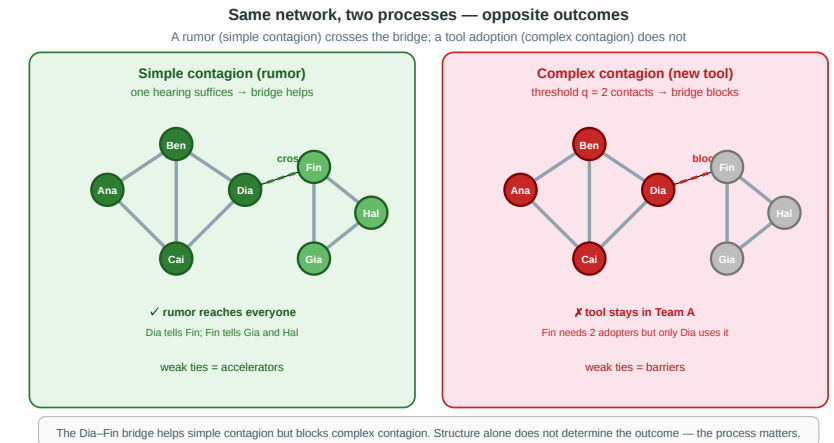
The Dia–Fin bridge helps simple contagion but blocks complex contagion. Structure alone does not determine the outcome — the process matters.

**Left:** A rumor (simple contagion) starts with Ana. One hearing suffices — it crosses the Dia–Fin bridge and reaches everyone. **Right:** A new tool (complex contagion, threshold  $q = 2$ ) starts with Ana. Fin needs 2 adopter contacts, but only Dia uses it — the bridge blocks adoption. Same edge, opposite effects.

# Four Questions from One Comparison

1. **Simple contagion.** If a disease spreads through any single contact, which structural features make it spread fastest?
2. **Complex contagion.** If a behavior requires reinforcement from multiple neighbors, how does the same structure help or hinder?
3. **Intervention.** Should we vaccinate the hub (Dia) or the peripheral node (Eli)? Does the answer change for disease vs. behavior?
4. **Timing.** If the Dia–Fin edge only exists on Tuesdays, does the rumor still cross? Does timing change the outcome?

Every concept in today's lecture answers one of these.



# A Sixth Kind of Gap

The course has accumulated five gaps between theory and data. L08 adds a sixth: the **process-structure interaction gap**.

| Lecture | Gap type                                                                               |
|---------|----------------------------------------------------------------------------------------|
| L03–L04 | Computational (NP-hard ideals vs. polynomial proxies)                                  |
| L05     | Causal (mechanism unidentifiable from snapshots)                                       |
| L06     | Structural (completeness required but absent)                                          |
| L07     | Navigational (paths exist but aren't findable locally)                                 |
| L08     | <b>Process-structure</b> (same structure, different outcomes depending on the process) |

# A Sixth Kind of Gap

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| L08     | <b>Process-structure</b> (same structure, different outcomes depending on the process) |

The process-structure gap is not about missing data or computational limits. It is about the fact that **network structure is necessary but not sufficient** — the spreading rule determines which structural features matter and how.



# Takeaway

The same edge can accelerate one process and block another. Network dynamics require knowing both the structure (the graph) and the process (the spreading rule) — neither alone determines the outcome.



# Simple Contagion and the SIR Model

**Guiding Question:** How do diseases and rumors spread through a network, and what structural features control the speed?

# Simple Contagion — Definition

**Simple contagion.** A node becomes active (infected, informed) through a **single** contact with an active neighbor. The probability of transmission per contact is independent of how many other active neighbors exist.

In network terms: each edge from an active to an inactive node carries independent transmission probability  $\beta$  per time step.

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**Key structural implication:** Weak ties and bridges *help* simple contagion — they carry the process to new communities that would otherwise be unreachable. This is why epidemics and rumors spread globally: they exploit the same inter-community shortcuts that Granovetter identified as carriers of novel information.

# The SIR Model

## The SIR model — compartments and transitions



Basic reproduction number:  $R_0 = \beta / \gamma \times \langle k \rangle$

Epidemic spreads if  $R_0 > 1$  — depends on both transmission rate and network degree

### SIR model on a network.

Each node is in one of three states: **Susceptible** (S), **Infected** (I), or **Recovered** (R).

- **Infection:** Each S-I edge transmits with probability  $\beta$  per time step.
- **Recovery:** Each I node recovers with probability  $\gamma$  per time step.
- **Basic reproduction number:**  $R_0 = \frac{\beta}{\gamma} \cdot \langle k \rangle$ , where  $\langle k \rangle$  is the average degree.

**Epidemic threshold:** The disease spreads (becomes an epidemic) if  $R_0 > 1$ .

# The Role of Hubs

In homogeneous networks, the average degree is often enough. In heterogeneous networks, the **variance of degree** matters: hubs create many more transmission opportunities than typical nodes.

For an uncorrelated random network, the epidemic threshold depends on the first two moments of the degree distribution:

$$T_c \approx \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}$$

Here  $T$  is transmissibility across an edge. If  $T > T_c$ , a macroscopic outbreak is possible.

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**Pastor-Satorras & Vespignani (2001).** In an idealized infinite scale-free network with  $P(k) \propto k^{-\gamma}$  and  $\gamma \leq 3$ ,  $\langle k^2 \rangle$  diverges, so  $T_c \rightarrow 0$ .

# Reading the Hub Result Correctly

## What the theorem says

- Hubs make simple contagion unusually robust.
- In an idealized infinite scale-free network with  $\gamma \leq 3$ , no finite epidemic threshold remains.
- Even low transmissibility can spread if it reaches sufficiently connected nodes.

## What it does *not* say

- Not every real disease infects everyone.
- Real contact networks are finite, layered, clustered, temporal, and capacity-limited.
- Immunity, vaccination, behavior change, geography, and pathogen biology restore practical thresholds.



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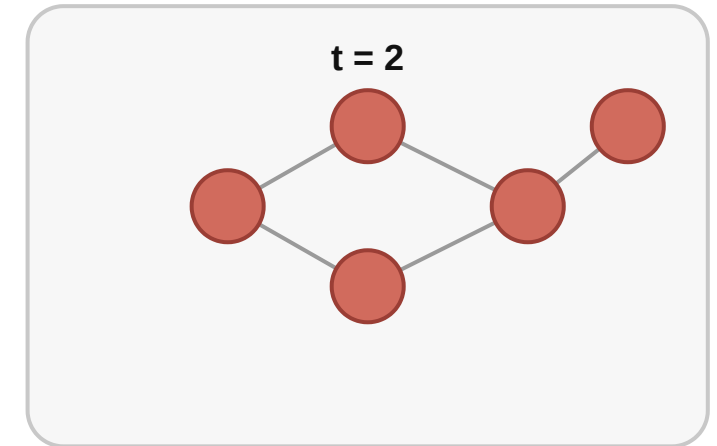
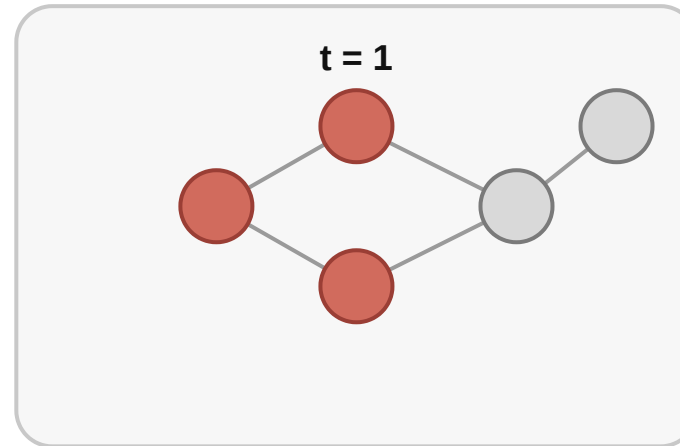
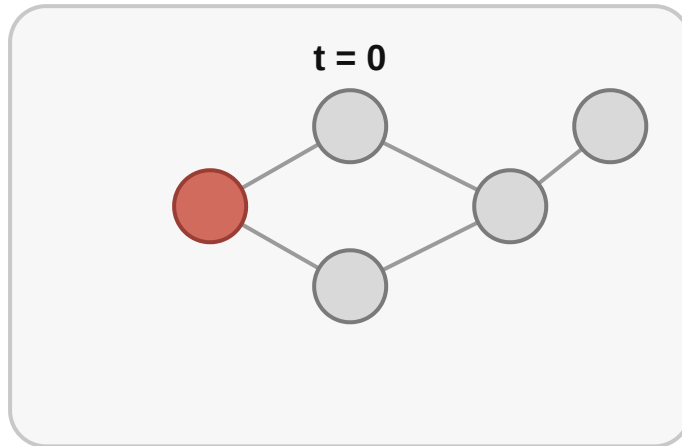
## What it does *not* say

- Not every real disease infects everyone.
- Real contact networks are finite, layered, clustered, temporal, and capacity-limited.
- Immunity, vaccination, behavior change, geography, and pathogen biology restore practical thresholds.

So the policy implication is not fatalism. It is **targeted intervention**: protect or reduce exposure around high-degree and high-betweenness nodes because they disproportionately sustain transmission.

# Diffusion on a Network

## Diffusion over time



Red nodes are active; each step spreads activity to reachable neighbors.

A simple contagion starting from a single node. At each step, each active node transmits to each inactive neighbor with probability  $\beta$ . Degree and centrality determine which nodes are reached first; communities determine where the wavefront pauses.

# Takeaway

Simple contagion spreads through any single contact. Weak ties and bridges accelerate global spread by carrying the process to new communities. Hubs are superspreaders. Vaccination strategies should target high-degree or high-betweenness nodes.

# Quick Check

In the workplace network, a cold virus starts with Ana ( $\beta = 0.3$  per contact per day,  $\gamma = 0.2$  per day). Ana has 4 direct contacts (Ben, Cai, Dia, Eli).

1. Estimate  $R_0$  using Ana's degree as a proxy for  $\langle k \rangle$ .
2. Will the cold become an epidemic in this workplace?
3. If you could vaccinate exactly one person to slow the spread, who should it be?

# Quick Check — Solution

$$R_0 = \frac{\beta}{\gamma} \times \langle k \rangle = \frac{0.3}{0.2} \times 4 = 6$$

Yes —  $R_0 = 6 \gg 1$ . The virus will spread through the workplace.

She is the only bridge to Team B. Removing her from the susceptible pool severs the only transmission path between teams — confining the epidemic to Team A. This is the dynamical version of "target the broker" from L04.

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3. **Vaccinate Dia.** She is the only bridge to Team B. Removing her from the susceptible pool severs the only transmission path between teams — confining the epidemic to Team A. This is the dynamical version of "target the broker" from L04.



# Complex Contagion and Threshold Cascades

**Guiding Question:** Why do some behaviors need social reinforcement — and how does this change which network structures help or hinder spreading?

# Complex Contagion — Definition

**Complex contagion.** A node adopts when at least a **fraction  $q$**  of its neighbors have already adopted. This models behaviors that require social reinforcement:

- adopting a new technology (need to see colleagues using it)
- joining a protest (need to see enough friends going)
- changing a social norm (need to see enough peers conforming)

**Threshold rule:** Node  $v$  adopts if  $\frac{|\text{active neighbors of } v|}{|\text{all neighbors of } v|} \geq q$ .

# The Weak-Tie Paradox

**In simple contagion:** weak ties and bridges *accelerate* spread — they carry the process to new communities. Granovetter's "strength of weak ties."

**In complex contagion:** weak ties and bridges *block* spread — a single cross-community contact cannot provide enough reinforcement to trigger adoption. The bridge is *too thin*.

# The Weak-Tie Paradox

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**In complex contagion:** weak ties and bridges *block* spread — a single cross-community contact cannot provide enough reinforcement to trigger adoption. The bridge is *too thin*.

**What helps complex contagion instead?** Dense clusters and wide bridges — multiple ties between communities that can jointly provide enough reinforcement to cross the threshold.

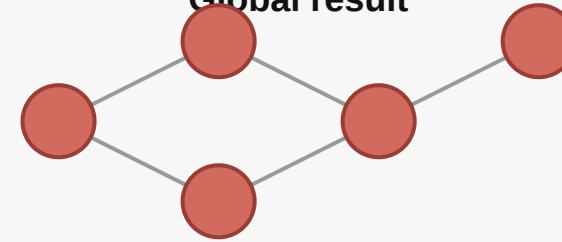
# Cascade Example

## Threshold cascade

### Local rule

activate  $v$  if  
active neighbors of  $v$   
-----  $\geq$   $\theta$   
total neighbors of  $v$

### Global result



A local adoption threshold can trigger a global cascade.

A threshold cascade starting from two initially active nodes. At each step, nodes whose active-neighbor fraction meets the threshold  $q$  adopt. The cascade may spread globally or stall — the outcome depends on both the threshold  $q$  and the local network density around the wavefront.

# When Do Cascades Go Global?

**Cascade condition.** In a network where each node has the same threshold  $q$ , a global cascade starting from a small seed set is possible if:

- the network contains enough **dense clusters** (where neighbors reinforce each other above  $q$ )
- these clusters are linked by **wide bridges** (multiple edges providing enough reinforcement to trigger adoption in the neighboring cluster)
- the threshold  $q$  is not too high (if  $q > 1/2$ , cascades require majority adoption locally, which is hard to trigger)

# Back to the Workplace

| Feature                     | Simple contagion (rumor)                 | Complex contagion (tool, $q = 2$ )          |
|-----------------------------|------------------------------------------|---------------------------------------------|
| Dia-Fin bridge              | Carries the rumor across                 | Blocks the tool — 1 contact < threshold     |
| Dense Team A core           | Helps locally, not critical for crossing | Essential — sustains internal adoption      |
| Strategy to spread globally | Protect the bridge                       | Widen the bridge (add more cross-team ties) |
| Vaccination / blocking      | Vaccinate Dia to stop spread             | No single node blocks it — density matters  |

# Takeaway

Simple contagion spreads through any single contact — weak ties help. Complex contagion requires social reinforcement — weak ties block. The same network structure plays opposite roles depending on the process, and intervention strategies must account for which kind of contagion is at work.



# Quick Check

A social network has two dense communities of 20 people each, connected by:

- **Scenario A:** a single bridge edge
- **Scenario B:** 5 parallel edges between the two communities

A behavior with threshold  $q = 0.2$  starts in community 1 (all 20 adopt).

1. In Scenario A, will the behavior cross to community 2?
2. In Scenario B, will it cross? Show your reasoning for a node in community 2 that has 10 neighbors total.
3. Would a *disease* (simple contagion) cross in both scenarios?

# Quick Check — Solution

The bridging node has ~1 active neighbor out of ~10 total:  $1/10 = 0.1 < q = 0.2$ . The single bridge is too thin.

A node with 5 cross-community ties:  $5/10 = 0.5 \geq q = 0.2$ . The wide bridge provides enough reinforcement.

Simple contagion needs only one S-I contact. Even Scenario A's single bridge suffices.

# Quick Check — Solution

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A node with 5 cross-community ties:  $5/10 = 0.5 \geq q = 0.2$ . The wide bridge provides enough reinforcement.

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# Quick Check — Solution

- 1. Scenario A: No.** The bridging node has ~1 active neighbor out of ~10 total:  $1/10 = 0.1 < q = 0.2$ . The single bridge is too thin.
- 2. Scenario B: Yes.** A node with 5 cross-community ties:  $5/10 = 0.5 \geq q = 0.2$ . The wide bridge provides enough reinforcement.

Simple contagion needs only one S-I contact. Even Scenario A's single bridge suffices.

# Quick Check — Solution

1. **Scenario A: No.** The bridging node has ~1 active neighbor out of ~10 total:  $1/10 = 0.1 < q = 0.2$ . The single bridge is too thin.
2. **Scenario B: Yes.** A node with 5 cross-community ties:  $5/10 = 0.5 \geq q = 0.2$ . The wide bridge provides enough reinforcement.
3. **A disease crosses in both.** Simple contagion needs only one S-I contact. Even Scenario A's single bridge suffices.

# Empirical Anchor: Centola's Online Experiment

**Guiding Question:** Does the simple/complex contagion distinction hold up empirically?

# The Study

Centola (2010), *The Spread of Behavior in an Online Social Network Experiment*, [Science 329, 1194–1197](#).

Centola created an online health forum and randomly assigned **1 528 participants** to one of two network structures:

| Condition | Structure                             | Clustering |
|-----------|---------------------------------------|------------|
| Clustered | Lattice-like (high clustering)        | High       |
| Random    | Watts-Strogatz-style (low clustering) | Low        |

Both networks had the same degree and diameter. Participants could register for a health program, and registration was visible to their network neighbors.

# What Centola Found

## 1. Behavior spread faster and further in the clustered network.

- In the clustered condition, adoption reached **54%** of the network.
- In the random condition, adoption reached only **38%**.

## 2. Adoption was driven by *multiple* exposures, not single contacts.

- Participants who saw 2+ neighbors register were far more likely to adopt than those who saw only 1 — consistent with a threshold rule.



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- Participants who saw 2+ neighbors register were far more likely to adopt than those who saw only 1 — consistent with a threshold rule.

## 3. The random network had shorter paths — so if simple contagion were the mechanism, it should have won. It lost. **Clustering beat shortcuts for behavior spreading.**

# What the Study Could Not See

- 1. Binary behavior.** The study tracked registration (yes/no), not the *intensity* or *quality* of adoption. A participant who registered but never used the program was counted the same as an active user.
- 2. Controlled networks  $\neq$  natural networks.** The assigned structures were regular and static. Real social networks are heterogeneous, dynamic, and have degree distributions that the experiment did not replicate.
- 3. Single behavior type.** The study tested one health-program registration. Whether the clustering advantage generalizes to other complex contagions (political mobilization, technology adoption, norm change) was not tested directly.
- 4. No decay or reversal.** Once registered, participants stayed registered. Real adoption can reverse — people abandon tools, lose interest, or face counter-influence.

# Takeaway

Centola (2010) experimentally confirmed the simple/complex contagion distinction: behavior spreading is faster in clustered networks (where local reinforcement is available), not random networks (where paths are shorter). Clustering beats shortcuts for complex contagion.

# Temporal Networks

**Guiding Question:** What do we miss when we replace an evolving network with a single static snapshot?

# Time-Respecting Paths

A **temporal network** is a graph where each edge  $e$  has an activation time  $t(e)$ .

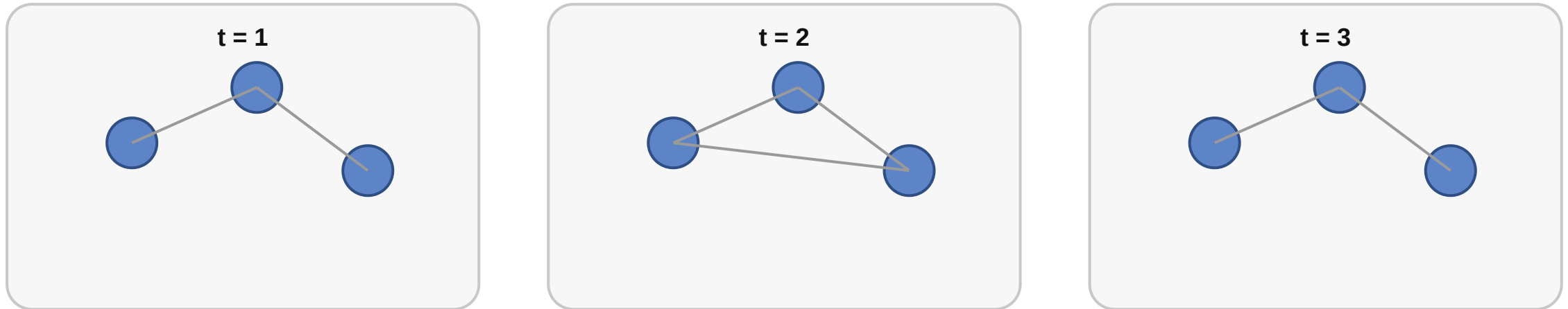
A **time-respecting path** from  $u$  to  $v$  is a sequence of edges  $e_1, e_2, \dots, e_k$  where:

- the edges form a path from  $u$  to  $v$
- $t(e_1) < t(e_2) < \dots < t(e_k)$

A path that exists in the *aggregated* static network may not exist as a time-respecting path — the edges may activate in the wrong order.

# Snapshot View

## Temporal network as changing snapshots



A static graph is one slice; temporal networks track when edges appear, disappear, or reorder.

The same set of nodes at three time points. An edge present in the aggregated static graph (taking the union across all snapshots) may not support a time-respecting path if the edges activate in the wrong order.

# Consequences for Spreading

1. **Reachability shrinks.** The set of node pairs connected by time-respecting paths is generally *smaller* than the set connected by static paths.
2. **Speed depends on timing.** A contagion may reach a node in 2 static hops but require waiting 5 time steps for edges to activate in the right order.
3. **Aggregation hides bottlenecks.** A bridge that exists only briefly may appear permanent in the aggregated graph, masking its role as a temporal bottleneck.

# Takeaway

Temporal networks make edge timing explicit. Static aggregation can hide bottlenecks and create phantom paths. For time-sensitive processes (epidemics, cascades, information flow), the order of edge activations matters as much as which edges exist.



# Quick Check

A temporal network has 4 nodes (A, B, C, D) and the following edges with activation times:

- A-B at  $t = 1$  1. Does a time-respecting path from A to D exist?
- B-C at  $t = 3$  2. Does a time-respecting path from A to D exist in the *static* aggregated graph?
- C-D at  $t = 2$  3. What is the minimum change needed to create a time-respecting A→D path?

# Quick Check — Solution

**1. No.**  $A-B (t = 1) \rightarrow B-C (t = 3)$  works, but  $C-D$  activates at  $t = 2$ , which is before the process reaches  $C$  at  $t = 3$ . The  $C-D$  edge has already "expired."

The aggregated static graph has path  $A-B-C-D$ . Static analysis hides the timing conflict.

(e.g.,  $t = 4$ ), giving the time-respecting path  $A-B (t = 1) \rightarrow B-C (t = 3) \rightarrow C-D (t = 4)$ .

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- 2. Yes.** The aggregated static graph has path  $A-B-C-D$ . Static analysis hides the timing conflict.
- 3. Change  $C-D$  to  $t > 3$**  (e.g.,  $t = 4$ ), giving the time-respecting path  $A-B (t = 1) \rightarrow B-C (t = 3) \rightarrow C-D (t = 4)$ .

# Wrap-Up — The Six Gaps

| Lecture | Gap type          | Core tension                                               |
|---------|-------------------|------------------------------------------------------------|
| L03–L04 | Computational     | Exact optimization is NP-hard                              |
| L05     | Causal            | Mechanism unidentifiable from snapshots                    |
| L06     | Structural        | Complete-graph theory meets sparse data                    |
| L07     | Navigational      | Short paths exist but aren't locally findable              |
| L08     | Process-structure | Same structure, different outcomes for different processes |
|         | Temporal          | Static aggregation hides causal order                      |

The course has built a single message: **network analysis requires matching the right structural concept to the right question**, and there is no single structural measure that answers every question. Edges, roles, communities, balance, navigability, and dynamics each require their own tools — and each reveals a different kind of gap between what we want to know and what the data can show.

[Reading](#)

Chapter 19 of Easley & Kleinberg (2010) covers cascades and thresholds. Chapter 21 covers epidemics. Centola (2010), *The Spread of Behavior in an Online Social Network Experiment* (Science), is the key empirical study. Holme & Saramäki (2012), *Temporal Networks* (Physics Reports), is the standard survey on time-varying graphs.